

Pangenome Analysis Reveals Gene Content Variation and Evolutionary Dynamics Across the *Culex pipiens* Species Complex

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Abstract

The *Culex pipiens* species complex contains the principal northern-hemisphere vectors of West Nile virus, lymphatic filariasis and several arboviruses; yet, the genomic basis of its diversity remains poorly resolved. We present the first pangenome characterization of the four widely recognized *Cx. pipiens* complex forms (*Cx. pipiens*, *Cx. pallens*, *Cx. quinquefasciatus*, *Cx. molestus*), with *Cx. tarsalis* as outgroup. From chromosome-scale assemblies (533–790 Mb) and unified annotations transferred from the *Cx. quinquefasciatus* reference, we identified 16,568 orthogroups. Restricted to the ingroup, 92,821 genes (98.2 %) were partitioned into 11,284 core (71.1 %), 3,726 shell (23.5 %) and 871 cloud (5.5 %) orthogroups. Form-specific cloud orthogroups were unevenly distributed ($\chi^2 = 71.9$, $p = 1.7 \times 10^{-15}$), with *Cx. molestus* carrying the largest set. A concatenated 8,067-locus phylogeny placed *Cx. pallens* and *Cx. quinquefasciatus* as sisters with full bootstrap support but only 58.9 % gene-tree and 45.1 % site concordance at the deep ingroup node, a pattern consistent with prior evidence that *Cx. pallens* arose through hybridization between *Cx. pipiens* and *Cx. quinquefasciatus*. CAFE5 detected 617 significantly evolving gene families, with all lineages showing strong expansion bias (*Cx. quinquefasciatus* greatest, 3,733/729). Whole-genome ANI ranged 93.1–94.7 % (skani); synteny was conserved at the chromosome scale (74–77 % collinear anchors) with 258–294 inversions ≥ 100 kb per pair. Transposable elements occupied a uniform genome fraction (52.7–54.4 %). These results refine the divergence framework for the complex, provide evidence consistent with a hybrid origin of *Cx. pallens*, and establish a pangenomic baseline for future population-level resequencing.

Keywords: *Culex pipiens*; species complex; pangenome; phylogenomics; gene family evolution; hybrid origin; West Nile virus

Highlights

- First chromosome-scale pangenome of the *Culex pipiens* species complex.

- 16,568 orthogroups partitioned into core, shell and cloud compartments.
- Phylogenomic discordance supports a hybrid origin for *Cx. pallens*.
- CAFE5 reveals strong, lineage-wide gene-family expansion across the complex.
- Conserved synteny and ~53% transposable-element content across all genomes.

1. Introduction

Mosquito-borne diseases remain among the most significant threats to global public health, with *Culex* mosquitoes serving as primary vectors for several neurotropic arboviruses and filarial nematodes (Farajollahi et al., 2011; Hayes et al., 2005). The *Culex pipiens* species complex, which includes *Cx. pipiens* (with recognized forms *pipiens* and *molestus*), *Cx. quinquefasciatus*, *Cx. pallens*, *Cx. australicus* and *Cx. globocoxitus*, occupies a near-cosmopolitan distribution and is responsible for the transmission of West Nile virus across temperate North America and Europe, St. Louis encephalitis virus, *Wuchereria bancrofti*, avian malaria and a range of additional arboviruses (Hayes et al., 2005; Farajollahi et al., 2011; Harbach, 2012).

Despite their close evolutionary relationships and the ability to produce fertile hybrids in laboratory and field settings, members of the complex exhibit pronounced ecological and behavioral differences that directly influence transmission dynamics. *Cx. p. pipiens* (the nominate form) is an anautogenous, ornithophilic, surface-breeding mosquito of temperate regions; *Cx. p. f. molestus* is autogenous, stenogamous, mammalophilic and exploits subterranean urban habitats; *Cx. quinquefasciatus* is the dominant tropical and subtropical sibling species and the principal urban *Culex* of the southern United States; and *Cx. p. pallens* is widely distributed across East Asia and shows a mixed bionomic profile (Vinogradova, 2000, 2003; Fonseca et al., 2004; Farajollahi et al., 2011). These differences have profound epidemiological consequences: the *molestus* form is implicated as a critical bridge vector for West Nile virus in temperate cities, while *Cx. quinquefasciatus* dominates lymphatic filariasis transmission across much of the tropics (Farajollahi et al., 2011).

The genomic basis for these phenotypic differences remains largely unresolved. Previous comparative studies have focused on individual species, pairwise contrasts, or population-level markers such as microsatellites, ACE-2, CQ11 and RADseq (Fonseca et al., 2004, 2009; Bahnck and Fonseca, 2006; Smith and Fonseca, 2004; Cornel et al., 2003; Kothera et al., 2009; Aardema et al., 2020, 2022). These studies have established that hybridization occurs wherever *Cx. p. pipiens* and *Cx. quinquefasciatus* are sympatric (Cornel et al., 2003; Kothera et al., 2009), that *Cx. p. pallens* carries a genomic signature of asymmetric, sex-linked introgression from *Cx. quinquefasciatus* into a *Cx. p. pipiens* background (Fonseca et al., 2009), and that across the complex *Cx. p. pallens* samples bear approximately 70 % *Cx. quinquefasciatus* and 30 % *Cx. pipiens / molestus* ancestry on a genome-wide RADseq scale (Aardema et al., 2020). What has been missing is a chromosome-scale, comparative-genomic framework that can resolve the relationships among the four widely recognized forms simultaneously and quantify the genomic compartments (core, shell and cloud orthogroups) that underlie their phenotypic divergence.

The pangenome concept, introduced for bacteria (Tettelin et al., 2005) and progressively extended to plants and animals (Golicz et al., 2020; Bayer et al., 2020), partitions the complete gene complement of a taxonomic group into a *core* shared by all members, a *shell* present in some but not all, and a *cloud* of taxon-restricted genes. Pangenomic analyses are common in microbial and crop genomics but remain uncommon in animal taxonomic groups; among mosquitoes, the most extensive multi-genome comparison to date is the 16-*Anopheles* analysis of Neafsey et al. (2015), which documented elevated rates of gene gain, loss and rearrangement relative to *Drosophila*. No analogous pangenomic resource has yet been published for the *Cx. pipiens* complex. Although the pangenome was originally defined for intraspecific variation, the core/shell/cloud framework is now routinely applied across clades and closely related species, including animal species groups (Eren et al., 2021; Golicz et al., 2020); it is in this clade-level sense that we use the term here.

Here we present the first pangenome analysis of the *Cx. pipiens* species complex. We integrate four chromosome-scale assemblies (*Cx. pipiens*, *Cx. pallens*, *Cx. quinquefasciatus*, *Cx. molestus*) with *Cx. tarsalis* as a deeper outgroup, applying a unified, fully reproducible Snakemake workflow that includes BUSCO assessment, annotation transfer, OrthoFinder-based orthology inference, core/shell/cloud pangenome partitioning, single-copy orthologue (SCO) concatenation phylogenomics with gene- and site-concordance assessment (IQ-TREE 2; Minh, Schmidt, et al., 2020; Minh, Hahn, et al., 2020), CAFE 5 gene-family-evolution modeling (Mendes et al., 2020), whole-genome ANI estimation by sketch-based and alignment-based methods (Shaw and Yu, 2023), pairwise synteny analysis and *de novo* transposable element annotation. We use the resulting framework to quantify pangenome architecture, to test the long-standing hypothesis of a hybrid origin for *Cx. pallens*, and to compare the *Cx. pipiens* complex to other medically important mosquito species complexes.

2. Materials and Methods

2.1 Genome assemblies

Five publicly available chromosome-scale genome assemblies were used: four members of the *Cx. pipiens* species complex and *Cx. tarsalis* as a deeper outgroup (Table 1). The four ingroup assemblies were retrieved from NCBI GenBank/RefSeq: *Cx. quinquefasciatus* (NCBI GCF_015732765.1), *Cx. pallens* (GCF_016801865.2), *Cx. molestus* (GCA_024516115.1) and *Cx. pipiens* (GCA_963924435.1). The *Cx. tarsalis* assembly CtarK1 (Main et al., 2021; deposited under the Open Science Framework project at osf.io/mdwqx) was used in place of the original NCBI accession GCA_016859205.1, which has been reassigned to an unrelated organism; the workflow therefore retrieves *Cx. tarsalis* from the OSF mirror as a special case.

Cx. quinquefasciatus GCF_015732765.1 was selected as the reference because it is the only assembly in the complex with a full NCBI RefSeq annotation, providing a curated set of gene models for downstream annotation transfer to the other ingroup forms and to the outgroup. Assembly contiguity and quality were assessed with QUAST v5.2.0 (Mikheenko et al., 2018)

and BUSCO v5 (Manni et al., 2021) using the *diptera_odb10* lineage dataset, in both genome and protein modes (Figure 1).

2.2 Annotation transfer

To eliminate annotation-source heterogeneity between assemblies, gene models were transferred from the *Cx. quinquefasciatus* RefSeq reference to the three remaining ingroup assemblies and to the *Cx. tarsalis* outgroup using Lifter v1.6.3 (Shumate and Salzberg, 2021) with default -copies -polish parameters. Lifter maps reference gene-region sequences with minimap2 (Li, 2018) and reconstitutes complete gene structures (mRNA, exon, CDS) on each target assembly. Protein sequences were extracted from each annotated genome with gffread v0.12.7 (Pertea and Pertea, 2020) using the -y flag to translate coding sequences from the transferred GFF3 annotations. The *Cx. quinquefasciatus* reference proteins were extracted directly from its RefSeq annotation.

The resulting protein sets totaled 24,531 (*Cx. quinquefasciatus*), 23,089 (*Cx. pallens*), 23,199 (*Cx. molestus*), 23,738 (*Cx. pipiens*) and 20,026 (*Cx. tarsalis*) sequences (Table 1). Protein-mode BUSCO completeness on the ingroup assemblies ranged from 77.9 % (*Cx. molestus*) to 95.3 % (*Cx. quinquefasciatus*); the *Cx. tarsalis* protein BUSCO score was lower (40.3 %), reflecting the expected loss of orthologue recovery when annotations are lifted across the deeper divergence between *Cx. tarsalis* and the reference (Figure 1).

2.3 Orthology inference and pangenome partitioning

Orthology was inferred with OrthoFinder v2.5.5 (Emms and Kelly, 2019) using DIAMOND v2.1.6 (Buchfink et al., 2015) for all-versus-all protein sequence comparison and Markov clustering (MCL) with an inflation parameter of 1.5. Orthogroups were partitioned post hoc into pangenome compartments restricted to the four ingroup forms: an orthogroup was assigned to the core if it contained at least one gene in every ingroup, to the shell if it contained genes in two or three of the four ingroups, and to the cloud if it contained genes in exactly one ingroup. Orthogroups in which all member genes derived from the *Cx. tarsalis* outgroup and no ingroup gene was present were classified as outgroup-only. Form-specific cloud orthogroups (cloud orthogroups restricted to a single named form) were tabulated separately. A χ^2 goodness-of-fit test was applied to the form-specific cloud counts to test the null hypothesis of equal cloud distribution among forms.

2.4 Phylogenomic analysis

Single-copy orthologues (SCOs) were defined as orthogroups containing exactly one gene per ingroup taxon. SCO protein sequences shorter than 50 amino acids were excluded as likely fragmentary calls. Each SCO was aligned individually with MAFFT v7.520 (Katoh and Standley, 2013) under default parameters, then trimmed with trimAl v1.4.1 using the -automated1 heuristic (Capella-Gutiérrez et al., 2009) to remove uninformative or saturated columns. After trimming,

alignments shorter than 50 amino acids were excluded; 8,067 trimmed alignments were concatenated into a supermatrix (3,868,063 amino acid sites, 0.007 % missing data) for concatenation-based maximum-likelihood inference. A species tree was inferred with IQ-TREE v2.2.6 (Minh, Schmidt, et al., 2020) using ModelFinder (-m MFP) for per-partition model selection and 1,000 ultrafast bootstrap replicates (-bb 1000). Gene concordance factors (gCF) and site concordance factors (sCF, 100 quartets) were computed against the per-locus gene trees and supermatrix sites respectively (Minh, Hahn, et al., 2020), providing complementary measures of phylogenetic signal beyond bootstrap support.

2.5 Gene family evolution

Gene family expansions and contractions across the species tree were modeled with CAFE 5 v5.1.0 (Mendes et al., 2020). Orthogroup gene-count tables from OrthoFinder were used as input. A gamma-distributed rate model with three rate categories ($k = 3$) was fitted, allowing different evolutionary rates to operate across gene families rather than enforcing a single uniform rate. The global birth–death rate parameter (λ) and the gamma shape parameter (α) were estimated by maximum likelihood; families with $p < 0.05$ for branch-specific rate departure were considered significantly evolving. Per-lineage expansion and contraction counts were tabulated and compared against a 1 : 1 null with a one-sided binomial test.

2.6 Whole-genome ANI

Pairwise whole-genome average nucleotide identity (ANI) across the five-taxon panel was estimated with skani v0.2 (Shaw and Yu, 2023), using the pyskani Python bindings with the default sketch parameters (compression = 125, $k = 15$). For the four chromosome-scale ingroup assemblies, sketching was restricted to the three longest scaffolds (~99 % of total assembly span); the contig-level *Cx. tarsalis* assembly was sketched in full. For all four *Cx. tarsalis*–ingroup pairs skani returned no estimate, including under permissive settings (cutoff = 0, learned_ani disabled, faster_small), indicating that genome-wide identity falls below the lower bound at which skani reports ANI (≈ 80 %); the four ingroup–ingroup pairs are therefore the only entries in the skani matrix. A supplementary alignment-based ANI was computed from minimap2 alignments (preset asm5/asm20; Li, 2018) as alignment-length-weighted mean identity ($\Sigma \text{ matches} / \Sigma (\text{matches} + \text{mismatches})$). For three of the six ingroup pairs alignment-based ANI was derived from the existing Liftoff intermediate alignments of the *Cx. quinquefasciatus* gene set onto each target (60–75 Mb of CDS per pair); for the remaining three pairs, partial chromosome-1 alignments (first ~2 Mb) were generated *de novo*. The skani matrix is reported as the primary deliverable; the alignment-based matrix is reported as a supplementary cross-check.

2.7 Pairwise synteny analysis

Pairwise large-scale synteny was assessed from the Liftoff gene-coordinate projections among the four ingroup taxa. For each pair, the intersection of gene IDs

shared between the two species (14,755–16,953 genes per pair) was used as anchors, with anchor coordinates taken as gene midpoints. For each query chromosome, the global slope of anchors against the dominant ortholog reference chromosome was computed; if negative (i.e., a whole-chromosome assembly-orientation mismatch rather than biology), the query y-coordinates of that chromosome were reflected so that the dominant diagonal was positive. Each anchor's local slope was computed over a window of five neighbors; anchors with positive local slope were classified as collinear (forward), and anchors with negative local slope were classified as inverted (reverse). Candidate inversions were enumerated as contiguous runs of ≥ 3 reverse-oriented anchors spanning ≥ 100 kb on the reference chromosome.

2.8 Transposable element annotation

De novo transposable element libraries were constructed for each genome with RepeatModeler v2.0.7 (Flynn et al., 2020). Species-specific libraries were then used to mask the corresponding genome with RepeatMasker v4.1.7 (Smit et al., 2013–2015). Repeat-class summary statistics (total bp masked, percent genome covered, interspersed repeats vs. simple/low-complexity) were extracted from the RepeatMasker .tbl output for each assembly.

All analyses were orchestrated as a single, end-to-end Snakemake workflow (Mölder et al., 2021) reproducible from the project repository at https://github.com/tylermaire/cx_pipiens_pangenome under Conda-managed software environments. The exact commit used to produce the results reported here is tagged in the repository. An overview of the full workflow is provided as Supplementary Figure S2.

3. Results

3.1 Genome assembly quality

The four ingroup chromosome-scale assemblies range in length from 533 Mb (*Cx. pipiens*) to 573 Mb (*Cx. quinquefasciatus*), with the *Cx. tarsalis* outgroup substantially larger (790 Mb; Table 1). Protein-mode BUSCO completeness against the *diptera_odb10* reference set was highest for the *Cx. quinquefasciatus* reference (95.3 %, of which 28.6 % duplicated) and ranged 77.9–82.2 % for the three Liftoff-annotated ingroup assemblies (Figure 1). The *Cx. tarsalis* protein-mode BUSCO score was 40.3 %, consistent with the expected loss of orthologue recovery when annotations are transferred across the deep divergence between *Cx. tarsalis* and the *Cx. pipiens* reference. Genome-mode BUSCO (which assesses the genome sequence directly rather than the annotation) was 79.6 % for *Cx. tarsalis*, confirming that the assembly itself is informative and that the lower protein-mode score reflects annotation-transfer attenuation rather than assembly fragmentation. Predicted protein counts (Table 1) ranged from 23,089 (*Cx. pallens*) to 24,531 (*Cx. quinquefasciatus*).

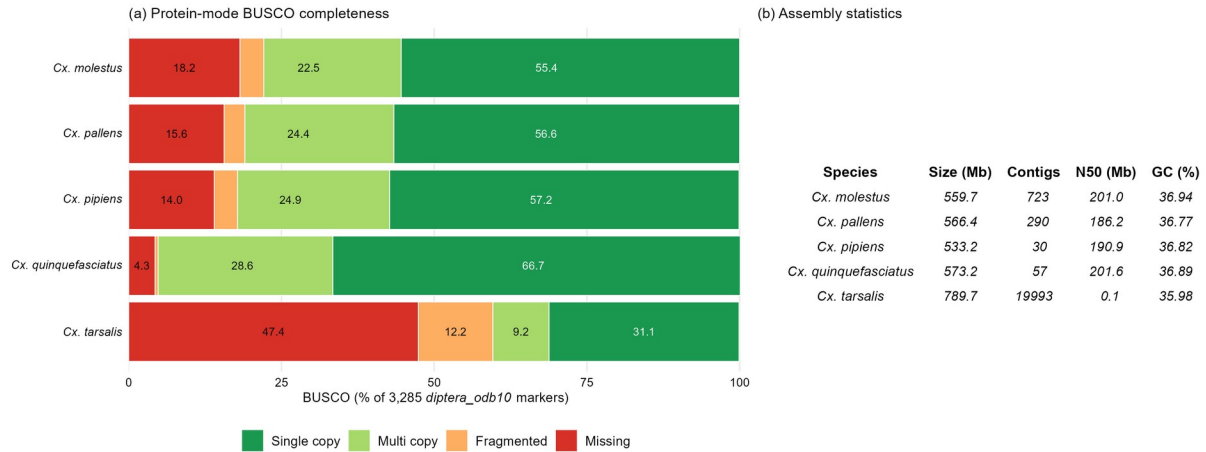


Figure 1. Genome assembly quality across the five-taxon panel. (a) Protein-mode BUSCO completeness against the *diptera_odb10* reference set (3,285 markers), partitioned into single-copy, multi-copy, fragmented and missing categories. (b) Assembly statistics from QUAST: total length, number of contigs, N50 and GC content.

Table 1. Genome assembly statistics and pangenome compartment occupancy for the *Cx. pipiens* species complex (four ingroup forms) and the *Cx. tarsalis* outgroup. Genome sizes are exact assembly totals (QUAST v5.2.0). Protein counts are the totals after annotation transfer with Liftoff v1.6.3 for the four non-reference assemblies, or the NCBI RefSeq protein set for *Cx. quinquefasciatus*. Per-form pangenome compartment columns give the number of genes in each ingroup form assigned to core, shell or cloud orthogroups by OrthoFinder v2.5.5. The *Cx. tarsalis* row is shown separately because its orthogroups are reported under the outgroup-only category, not core/shell/cloud.

Genome	Accession	Size (Mb)	Proteins	Core genes	Shell genes	Cloud genes
<i>Cx. molestus</i>	GCA_024516115.1	560	23,199	18,476	3,332	912
<i>Cx. pallens</i>	GCF_016801865.2	566	23,089	18,450	3,661	546
<i>Cx. pipiens</i>	GCA_963924435.1	533	23,738	18,596	4,052	645
<i>Cx. quinquefasciatus</i>	GCF_015732765.1	573	24,531	19,439	4,436	277
<i>Cx. tarsalis</i> (outgroup)	OSF CtarK1 mirror	790	20,026	—	—	—

3.2 Pangenome architecture

OrthoFinder identified 16,568 orthogroups across the five-taxon panel. Restricted to the four ingroup forms, 92,821 genes (98.2 % of ingroup gene calls) were assigned to 15,881 orthogroups, partitioned as 11,284 core (71.1 % of ingroup orthogroups), 3,726 shell (23.5 %) and 871 cloud (5.5 %). An additional 687 orthogroups were specific to the *Cx. tarsalis* outgroup

and contained no ingroup gene. Per-species totals (Table 1) ranged from 22,720 partitioned genes in *Cx. molestus* to 24,150 in *Cx. quinquefasciatus*; of the 11,284 core orthogroups, almost all (11,275 per form on average) contained at least one gene in every form, as expected for a strictly defined core, whereas shell and cloud occupancy varied substantially. The per-form core gene totals in Table 1 (18,450–19,439) exceed the core orthogroup count (11,284) because many core orthogroups contain more than one gene per form. The visual partition (Figure 2a) and per-form stacked composition (Figure 2b) show that the dominant signal across all four forms is a large, shared core, with form-to-form differences concentrated in the shell and cloud compartments.

Form-specific cloud orthogroups (cloud orthogroups containing genes from a single named form only) totalled 297 in *Cx. molestus*, 245 in *Cx. pipiens*, 203 in *Cx. pallens* and 126 in *Cx. quinquefasciatus* (Figure 2c). This distribution departed significantly from uniform ($\chi^2 = 71.9$, $df = 3$, $p = 1.7 \times 10^{-15}$). *Cx. molestus* carried the largest set of form-restricted cloud orthogroups despite having an intermediate total protein count, suggesting that the autogenous, hypogean *molestus* form has accumulated a distinct complement of lineage-restricted genes; we return to this in the Discussion.

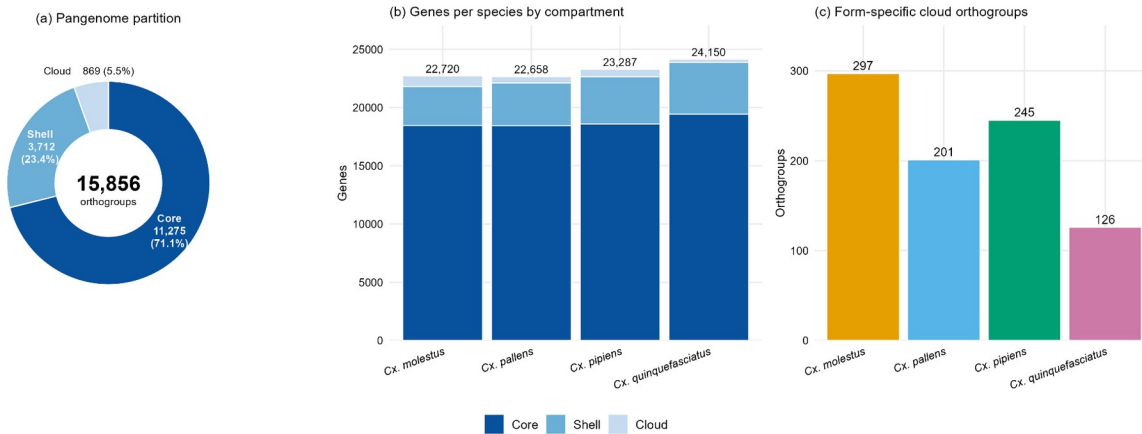


Figure 2. Pangenome composition of the four ingroup *Cx. pipiens* complex forms. (a) Donut chart of the ingroup pangenome partition into core, shell and cloud orthogroups. (b) Per-form stacked gene counts in each pangenome compartment. (c) Form-specific cloud orthogroups (cloud orthogroups containing genes from a single named form only). Statistical test of cloud-distribution uniformity (χ^2 goodness-of-fit) is reported in Section 3.2.

3.3 Phylogenomic relationships

Concatenation maximum-likelihood inference on the 8,067-locus supermatrix (3.87 Mb amino acid sites, 0.007 % missing data) produced a fully resolved species tree with full ultrafast bootstrap support at every internal branch (Figure 3). The recovered topology places *Cx. pallens* and *Cx. quinquefasciatus* as sister taxa, with *Cx. pipiens* as the next-closest relative and *Cx. molestus* as the outermost ingroup. The single internal ingroup branch separating (*pallens* + *quinquefasciatus*) from (*pipiens* + *molestus*) received 100 % bootstrap but substantially lower

concordance support: gCF = 58.9 % (3,991 of 6,774 decisive gene trees concordant) and sCF = 45.1 % (averaged over 100 quartets; Figure 3). Of the discordant gene trees, 18.2 % supported the first alternative NNI quartet and 22.9 % supported the second; sites mirrored this pattern with 28.4 % supporting one alternative and 26.5 % the other. Branch lengths within the ingroup were short (0.026–0.073 substitutions per site), consistent with relatively recent radiations.

The combination of full bootstrap support and substantially reduced concordance is not artefactual: it indicates that while the concatenated supermatrix strongly favors a single topology, the underlying gene trees and sites are split between alternative resolutions at this node. This pattern is the one expected when a lineage has a hybrid origin or has experienced substantial post-divergence introgression (Degnan and Rosenberg, 2009; Minh, Hahn, et al., 2020; Hibbins and Hahn, 2022), and we develop this interpretation in the Discussion.

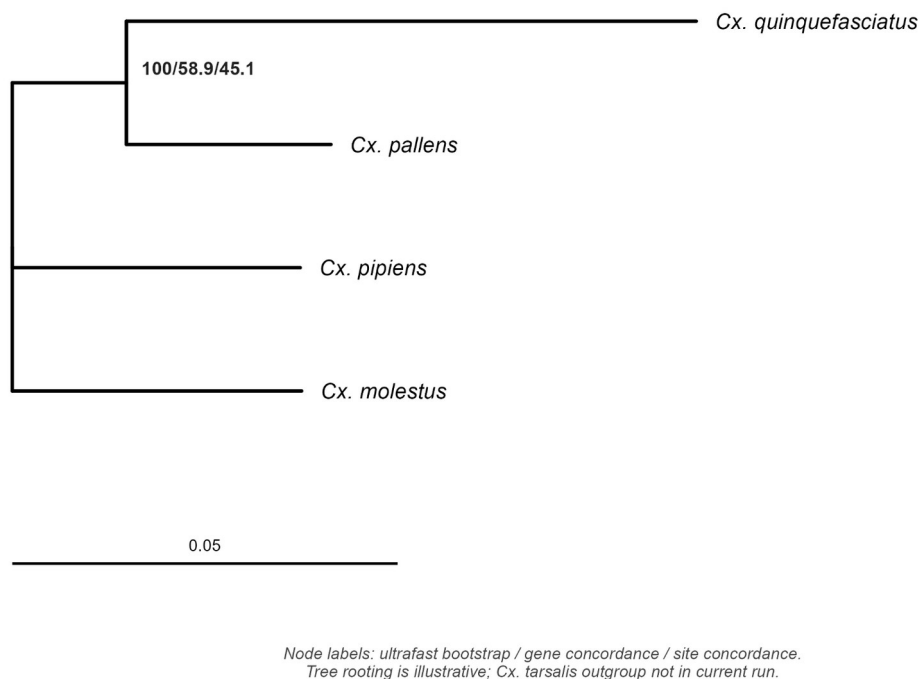


Figure 3. Concatenation maximum-likelihood species tree of the four *Cx. pipiens* complex ingroup taxa, inferred from 8,067 trimmed single-copy orthologue alignments (3,868,063 amino acid sites) with IQ-TREE 2 under ModelFinder-selected per-partition substitution models and 1,000 ultrafast bootstrap replicates. Node labels show ultrafast bootstrap support / gene concordance factor (gCF) / site concordance factor (sCF). Scale bar in amino acid substitutions per site. The tree is shown unrooted.

3.4 Gene family evolution

CAFE 5 identified 617 gene families showing significant rate departures across the species tree ($p < 0.05$) under a gamma model with three rate categories (Figure 4). The estimated birth-

death rate was $\lambda = 0.105$ substitutions per gene per time unit; the gamma shape parameter $\alpha = 0.167$ indicates substantial heterogeneity of evolutionary rates among families. Across all four ingroup lineages, expansion events outnumbered contractions by a factor of two to five (Figure 4), and the excess was significant on every branch (one-sided binomial test, $p < 2 \times 10^{-16}$ in every comparison). *Cx. quinquefasciatus* showed the strongest expansion bias (3,733 expanded families vs. 729 contracted, ratio 5.1), followed by *Cx. pipiens* (3,363 vs. 801, ratio 4.2), *Cx. pallens* (3,281 vs. 1,085, ratio 3.0) and *Cx. molestus* (3,220 vs. 1,263, ratio 2.5). The lineage ranking by expansion bias mirrors the order of lineage protein-count totals (Table 1) and is consistent with the general pattern of accelerated gene-family turnover reported across culicine mosquitoes (Arensburger et al., 2010; Neafsey et al., 2015).

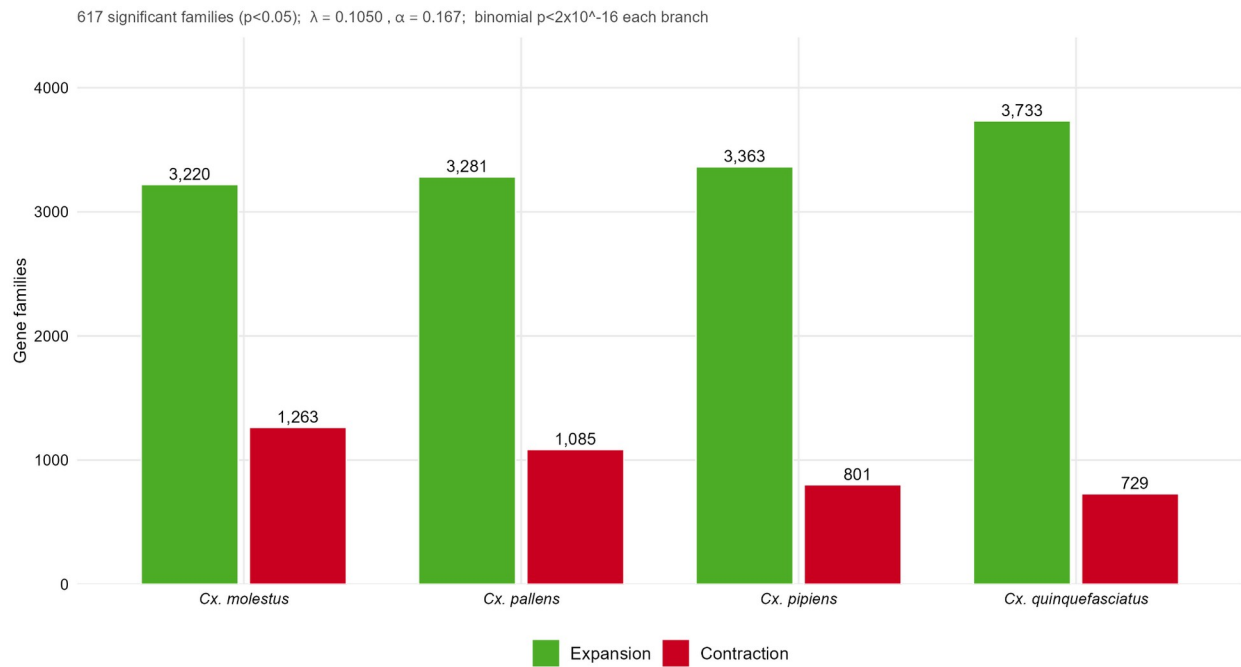


Figure 4. Per-lineage gene family expansion (green) and contraction (red) counts inferred by CAFE 5 under a gamma-distributed rate model with three rate categories ($k = 3$). Model parameters: $\lambda = 0.105$, $\alpha = 0.167$; 617 families significantly evolving at $p < 0.05$. Subtitle reports per-branch binomial-test results for the expansion-versus-contraction excess.

3.5 Whole-genome ANI and pairwise synteny

Pairwise whole-genome ANI (skani) across the four ingroup taxa ranged from 93.07 % (*Cx. pipiens* vs. *Cx. quinquefasciatus*) to 94.72 % (*Cx. molestus* vs. *Cx. pipiens*), with a mean of 93.87 % (Figure 5; Table 2). Alignment-based ANI on the same pairs (computed from minimap2 alignments of either the Liftoff-projected gene set or partial chromosome-1 alignments) was 1.5–2.0 percentage points higher, in the 95.09–96.57 % range, reflecting the expected bias of alignment-based metrics toward the conserved, alignable fraction of these repeat-rich genomes (~50 % TE content; Section 3.6). Both metrics agree on the qualitative pattern: the four ingroup forms are highly similar but not interchangeable at the nucleotide level, with the closest pair

being *Cx. molestus* and *Cx. pipiens* and the most divergent pair being *Cx. pipiens* and *Cx. quinquefasciatus*. *Cx. tarsalis* was included in the analysis but could not be placed on the same ANI scale as the ingroup: skani returned no estimate for any *Cx. tarsalis*–ingroup pair under default or permissive (cutoff = 0, learned_ani disabled, faster_small) settings, because the genome-to-genome identity falls below the lower bound at which skani reports reliable ANI ($\approx 80\%$). This absence of a sketch-based estimate is itself consistent with the substantial divergence expected for *Cx. tarsalis* as a deeper outgroup, and places its whole-genome identity to the ingroup well below the 93–95 % observed among the four forms.

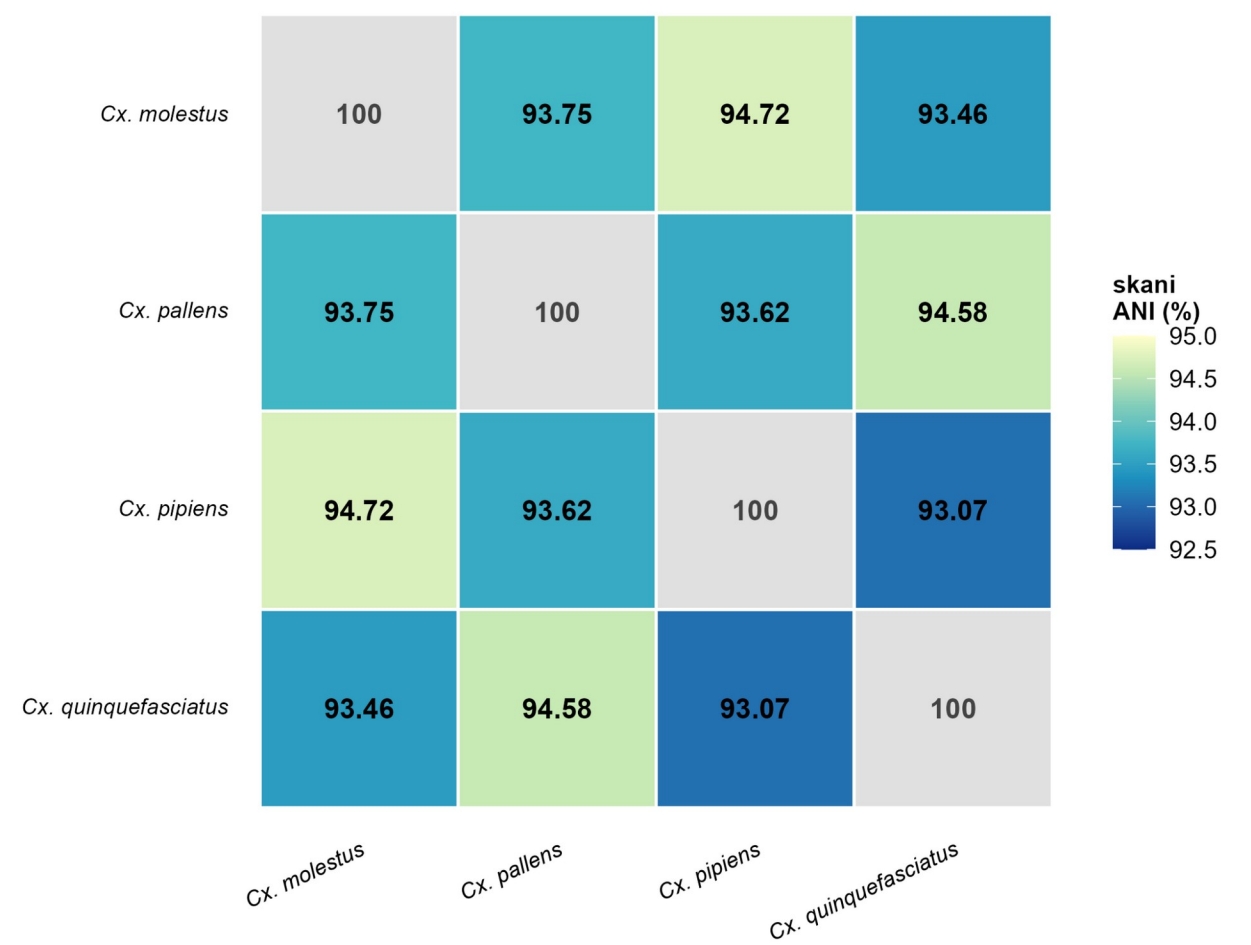


Figure 5. Pairwise whole-genome average nucleotide identity among the four ingroup forms, computed with skani v0.2 (default sketch parameters; compression 125, k = 15) restricted to the three chromosome-scale scaffolds of each assembly. Cell values are percent identity; diagonal cells are shown as 100 % on a neutral background.

Table 2. Pairwise whole-genome average nucleotide identity (%) among the four ingroup *Cx. pipiens* complex forms, computed with skani v0.2 (Shaw and Yu, 2023). *Cx. tarsalis* is listed as a separate row with no value (not estimable by skani; see Methods Section 2.6).

	Cx.	Cx.	Cx.	Cx.
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	<i>molestus</i>	<i>pallens</i>	<i>pipiens</i>	<i>quinquefasciatus</i>
<i>Cx. molestus</i>	100.00	93.75	94.72	93.46
<i>Cx. pallens</i>	93.75	100.00	93.62	94.58
<i>Cx. pipiens</i>	94.72	93.62	100.00	93.07
<i>Cx. quinquefasciatus</i>	93.46	94.58	93.07	100.00

Chromosome-scale synteny was preserved 1:1 across all six ingroup pairs (chr1 ↔ chr1, chr2 ↔ chr2, chr3 ↔ chr3). The fraction of shared gene anchors collinear with the dominant chromosome orientation ranged from 73.93 % (*Cx. molestus* vs. *Cx. quinquefasciatus*) to 77.10 % (*Cx. pipiens* vs. *Cx. quinquefasciatus*; Figure 6). Across all six pairwise comparisons we identified 258–294 candidate inversions ≥ 100 kb per pair; the largest single inversion (3.01 Mb) spans chromosome 3 between coordinates 72.25 Mb and 75.26 Mb of *Cx. pallens* relative to *Cx. pipiens*. Many of the smaller candidate inversions are concentrated in the repeat-rich centromeric regions of chromosome 2 in every pair and likely combine bona fide micro-inversions with annotation-projection noise; the larger blocks (≥ 1 Mb) recur across pairs and are stronger candidates for biologically meaningful rearrangements. The full per-pair synteny dotplots are provided as Supplementary Figure S1.

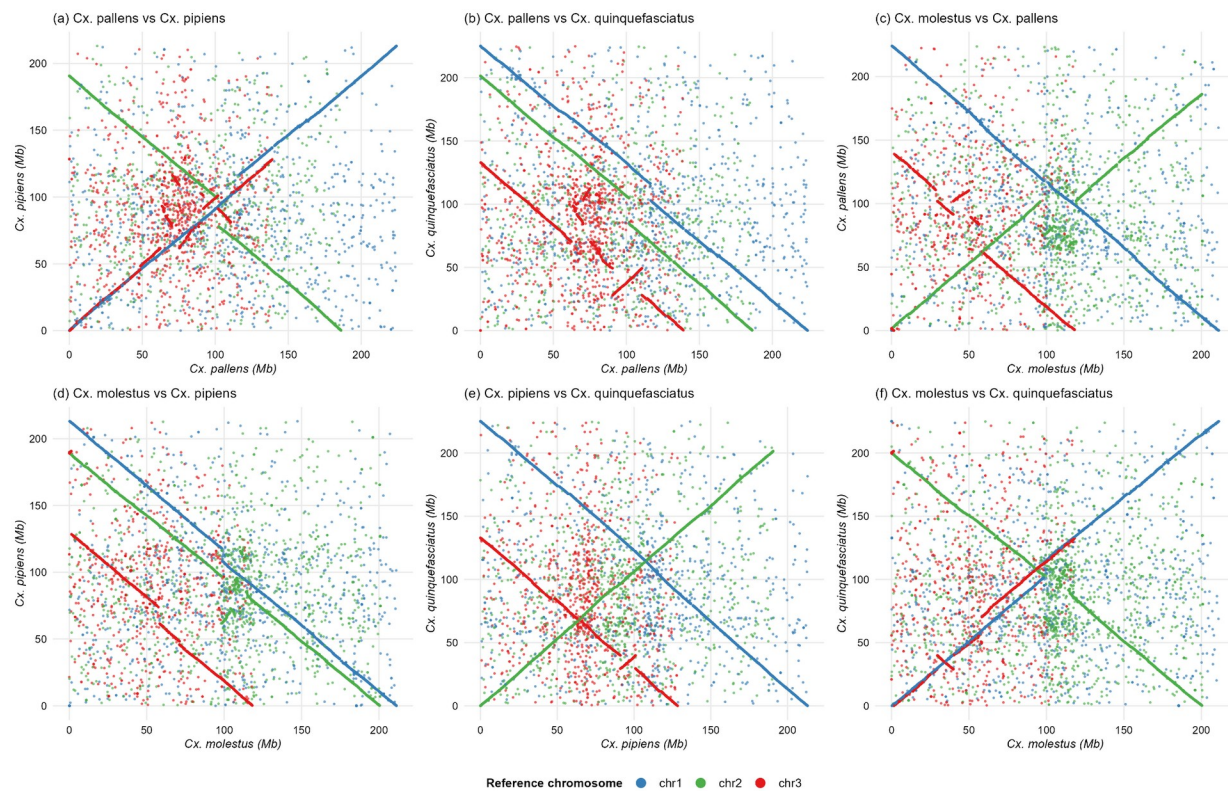


Figure 6. Pairwise gene-anchor synteny dotplots for the six ingroup genome comparisons. Each panel plots the genomic position (Mb) of every shared gene anchor on the reference genome (x-axis) against its position on the query genome (y-axis); points are colored by the reference genome's chromosome of origin (blue = chr1, green = chr2, red = chr3, ordered by size). Built in R with ggplot2 from the Liftoff-transferred gene coordinates; quantitative collinearity and inversion statistics are reported in Section 3.5.

3.6 Transposable element composition

Repeat-content analysis revealed uniform transposable element landscapes across the four ingroup genomes (Figure 7). Interspersed repeats accounted for 52.69 % (*Cx. pipiens*) to 54.43 % (*Cx. pallens*) of total assembly length, with *Cx. molestus* (53.46 %) and *Cx. quinquefasciatus* (53.86 %) intermediate. Simple and low-complexity repeats added a further ~3 % per genome. Total repeat content thus occupies more than half of each chromosome-scale assembly, in line with prior reports for culicine mosquitoes (Arensburger et al., 2010). The narrow inter-form variation in repeat-class proportions stands in marked contrast to the more variable protein counts (23,089–24,531; Table 1) and pangenome compartment occupancy reported above.

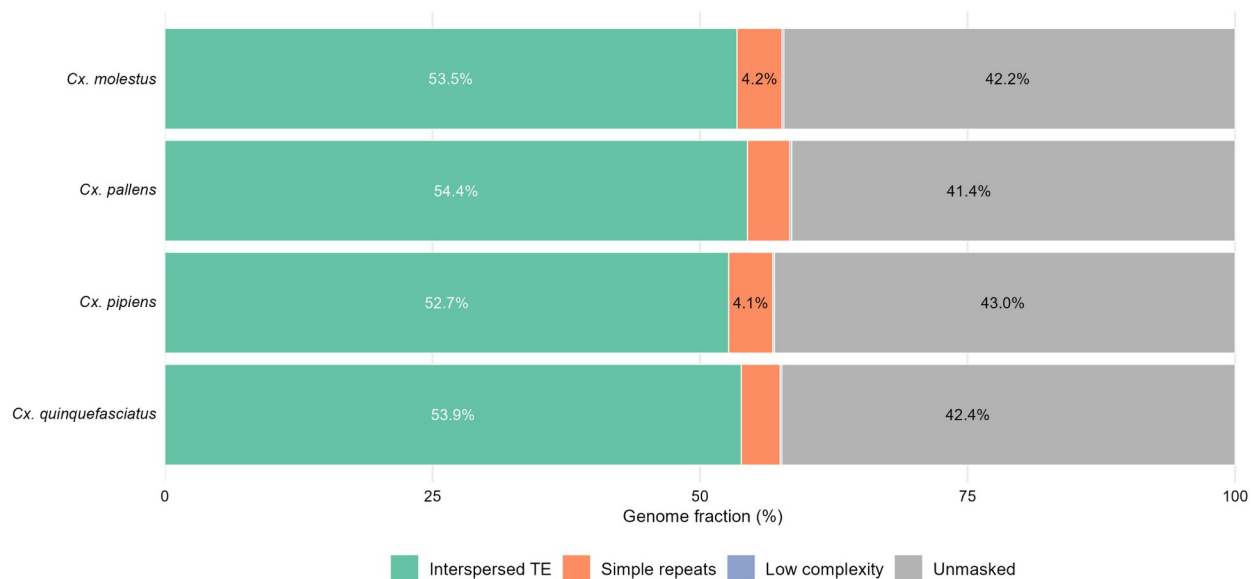


Figure 7. Transposable element composition across the four ingroup genomes. Stacked horizontal bars show the percent of each genome occupied by interspersed transposable elements, simple repeats, low-complexity sequences and unmasked sequence, as reported by RepeatMasker v4.1.7 run with species-specific de novo libraries built by RepeatModeler2.

3.7 Anvi'o gene-cluster pangenome

As an independent, finer-grained view of the pangenome, the four ingroup contigs databases (built with the same Liftoff-transferred gene calls used throughout this study) were assembled into a single anvi'o pangenome (anvi'o v9; Eren et al., 2021) using DIAMOND v2.1.6 for all-

versus-all protein comparison and Markov clustering with an inflation parameter of 10 and a minimum bit-score ratio (minbit) of 0.5. This anvi'o analysis was run manually outside the Snakemake workflow; the exact commands are documented in the project repository. Using one representative protein per gene (the longest isoform) yielded 57,815 input proteins across the four genomes and 16,673 gene clusters in the resulting pangenome (Figure 8). The total cluster count is close to the 15,881 ingroup orthogroups recovered by OrthoFinder (Section 3.2), indicating that the two clustering approaches converge on a similar partition when given the same input protein set, despite anvi'o's higher MCL inflation.

The intersection structure (Figure 8a) recovers the qualitative core/shell/cloud architecture described in Section 3.2: 9,423 clusters (56.5 %) are present in all four genomes (the “anvi'o core”), followed by mid-frequency shared sets (852 absent only from *Cx. molestus*; 713 absent only from *Cx. pallens*; 512 absent only from *Cx. pipiens*; 427 absent only from *Cx. quinquefasciatus*) and a long tail of singleton clusters present in only one genome (918 in *Cx. molestus*, 820 in *Cx. pipiens*, 708 in *Cx. pallens*, 613 in *Cx. quinquefasciatus*). Pairwise-only intersections account for an additional 1,687 clusters distributed across the six possible genome pairs. Per-genome totals range from 11,540 (*Cx. molestus*) to 11,860 (*Cx. quinquefasciatus*) clusters (Figure 8b), and the cluster-size distribution (Figure 8c) is dominated by small clusters (median 3 genes per cluster) with a long right tail of multi-copy gene families. The anvi'o singleton counts are consistent in rank order with the OrthoFinder form-specific cloud counts reported in Section 3.2 (*Cx. molestus* > *Cx. pipiens* > *Cx. pallens* > *Cx. quinquefasciatus*), providing methodological cross-validation for the conclusion that *Cx. molestus* harbors the most form-restricted genes of any form, even though its overall protein count is only intermediate.

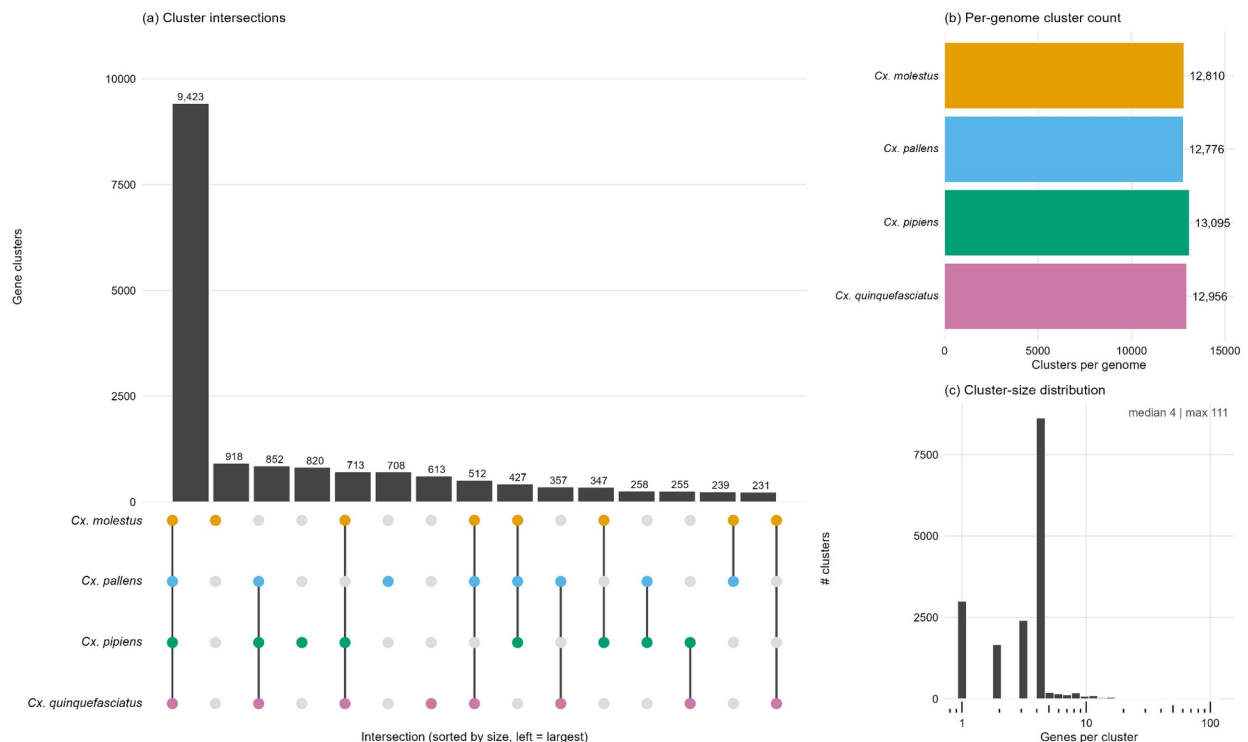


Figure 8. Anvi'o pangenome composition for the four *Cx. pipiens* complex ingroup genomes. (a) UpSet-style plot of gene-cluster intersections: each bar gives the number of clusters present in the genome subset indicated by the filled dots in the matrix below, with subsets ordered by intersection size. (b) Total gene clusters per genome. (c) Cluster-size distribution (genes per cluster, log scale; median and maximum annotated). Built with anvi'o v9 (Eren et al., 2021) using DIAMOND for all-versus-all protein comparison and MCL clustering at inflation 10 and minbit 0.5, on the Liftoff-transferred gene calls used throughout this study (one longest-isoform protein per gene, 57,815 input proteins).

3.8 Key gene families

To probe whether functional categories with documented relevance to vector biology vary in copy number across the complex, we examined orthogroup membership in three curated category sets covering detoxification (cytochrome P450, glutathione-S-transferase, esterase), chemosensory (odorant-binding proteins, chemosensory proteins, odorant receptors, gustatory receptors) and immunity (Toll, IMD, JAK/STAT pathway components, defensins, cecropins, attacins; categories profiled in depth for *Cx. quinquefasciatus* by Bartholomay et al., 2010), drawing on gene-name keyword matches to the Liftoff-transferred annotations. Per-form copy-number summaries are visualised as a heatmap in Figure 9. Across the complex, copy-number profiles are broadly conserved, with the largest inter-form differences concentrated in detoxification families and odorant receptors, categories that are repeated targets of selection in mosquito vectors. The full per-family per-form copy-number table is provided as Supplementary Table S6. The downstream anvi'o pangenomic visualization (Section 3.7) provides an orthogonal, gene-cluster-level confirmation that *Cx. molestus* has the most form-specific gene clusters, echoing the OrthoFinder partition at the gene-cluster level.

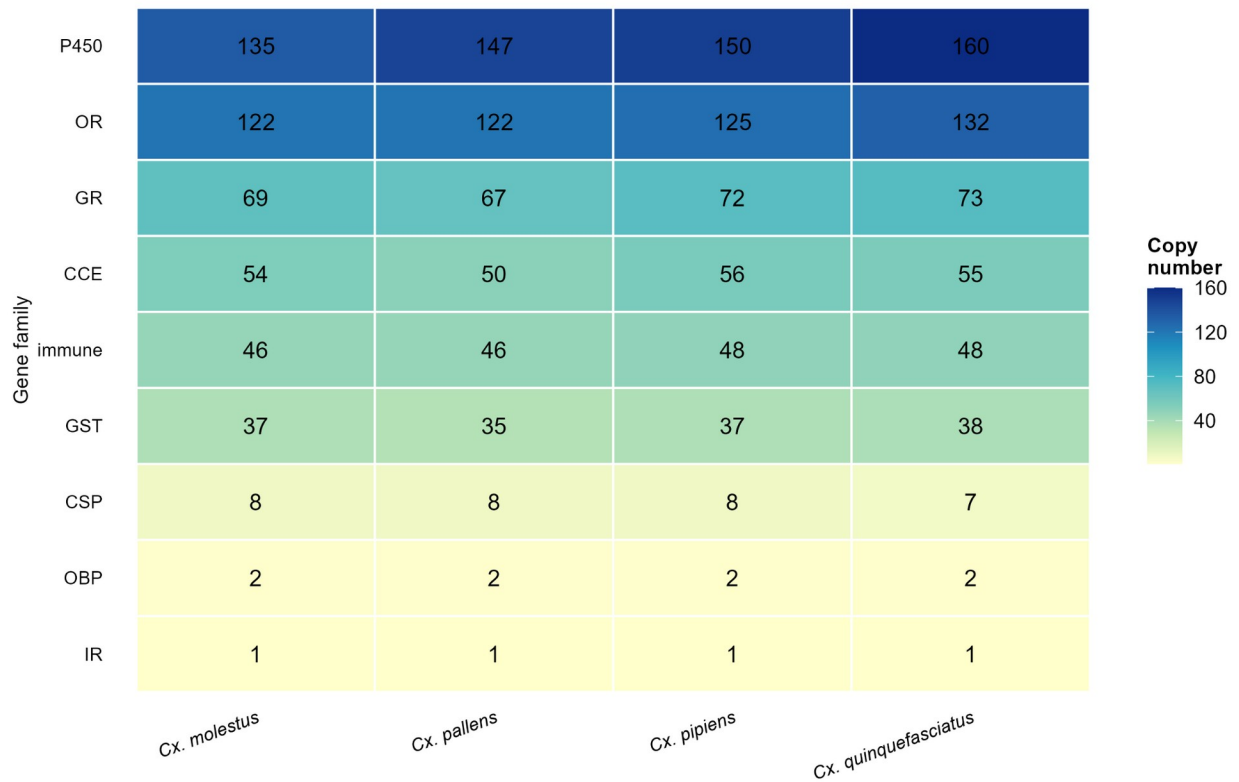


Figure 9. Per-form copy-number heatmap for three curated gene category sets covering detoxification (cytochrome P450, glutathione-S-transferase, esterase), chemosensory (odorant-binding proteins, chemosensory proteins, odorant and gustatory receptors) and immunity (Toll, IMD, JAK/STAT, antimicrobial peptide families). Cells show per-form orthogroup-level copy counts; rows are ordered by mean copy number across the four forms (highest on top). Per-family per-form table provided as Supplementary Table S6.

4. Discussion

We present the first chromosome-scale pangenome analysis of the *Cx. pipiens* species complex, integrating four ingroup assemblies and a *Cx. tarsalis* outgroup under a single, reproducible workflow. Several findings refine the existing genomic picture of this group, and one, the topology and discordance signal, provides genomic support for a long-standing biological hypothesis.

4.1 Rationale for a four-genome design

A four-genome panel is, by the standards of bacterial or plant pangenomics, a small design. We adopted it deliberately rather than expansively, for three reasons that bear on the interpretation of the results. First, the panel includes every chromosome-scale, publicly available assembly representing each of the four widely recognized *Cx. pipiens* complex forms; expanding to

additional taxa would require either (i) using fragmented draft assemblies whose contiguity precludes synteny analysis or (ii) waiting for additional reference-quality assemblies that are not yet available. Second, the addition of *Cx. tarsalis* as outgroup makes the orthology partition rootable while contributing minimal noise to the deep-node ingroup analyses, because its substantial divergence prevents it from participating in the SCO concatenation phylogeny at high resolution. Third, restricting the analysis to a single high-quality assembly per form keeps within-form polymorphism out of the shell and cloud compartments, which therefore reflect between-form divergence rather than population-level variation. The two omitted complex members, *Cx. australicus* and *Cx. globocoxitus*, are Australian endemics that sit as a monophyletic group sister to *Cx. quinquefasciatus* in published phylogenies, with a deeper divergence than the splits among the four forms analyzed here (Aardema et al., 2020); their inclusion would shift the rooting question to a more peripheral part of the complex without adding resolution to the central *pipiens/quinquefasciatus/pallens/molestus* radiation.

4.2 Phylogenomic discordance is consistent with a hybrid origin for *Cx. pallens*

The combination of a fully bootstrap-supported topology that places *Cx. pallens* and *Cx. quinquefasciatus* as sisters with substantially reduced gene- and site-concordance (gCF = 58.9 %, sCF = 45.1 %) at first appears contradictory. Concatenation maximum likelihood can produce inflated bootstrap support even when the underlying loci disagree (Minh, Hahn, et al., 2020); the gCF and sCF metrics were developed precisely to expose this kind of disagreement. Pure incomplete lineage sorting (ILS) is one source of such disagreement and is expected when speciation events occur in rapid succession (Degnan and Rosenberg, 2009). Hybridization and post-divergence introgression are another, and produce discordance with characteristic asymmetries, for example near-balanced support for the two alternative quartets when introgression is recent and pervasive (Hibbins and Hahn, 2022). In our data, the two NNI alternatives at the deep ingroup node received 18.2 % and 22.9 % gene-tree support respectively, and 28.4 % and 26.5 % site support, a near-balanced asymmetry suggestive of a process beyond pure ILS.

This pattern is not new in the *Cx. pipiens* complex; what is new is its expression at the whole-genome, chromosome-scale level. Fonseca et al. (2009) used ACE-2 intron sequences and eight microsatellite loci across Japanese, Korean and Chinese *Cx. p. pallens* populations to show that every sampled male carried both “pallens” and “quinquefasciatus” ACE-2 bands, while females overwhelmingly carried only the “pallens” form, explicitly evidencing asymmetric, sex-linked introgression of *Cx. quinquefasciatus* into *Cx. p. pallens* and concluding that *Cx. p. pallens* “may have arisen through hybridization between *Cx. pipiens* and *Cx. quinquefasciatus*” (Fonseca et al., 2009). Aardema et al. (2020) recovered the same signal at a genome-wide RADseq scale, reporting that *Cx. p. pallens* samples carry approximately 70 % *Cx. quinquefasciatus* and 30 % *Cx. pipiens/molestus* ancestry along with a substantial complement of private variants, consistent with a hybrid origin followed by independent evolution. Earlier work documented that *Cx. p. pipiens* and *Cx. quinquefasciatus* hybridise extensively wherever they are sympatric (Cornel et al., 2003; Kothera et al., 2009), establishing the biological pre-

conditions for such a hybrid lineage. Our phylogenomic recovery of *Cx. pallens* as sister to *Cx. quinquefasciatus*, rather than to *Cx. pipiens* as some earlier analyses based on fewer loci had suggested, is consistent with the dominant ~70 % *quinquefasciatus* component of the *pallens* ancestry quantified by Aardema et al. (2020), and the depressed concordance factors at the deep ingroup node are the pattern expected when a substantial fraction of the genome reflects an alternative history. These results therefore corroborate, at chromosome-scale resolution, a hybrid-origin model for *Cx. pallens* that has been progressively constructed from population-genetic markers over two decades.

4.3 Comparison with other insect vector complexes

Our analytical framework parallels, but does not duplicate, recent work on the *Anopheles gambiae* species complex. Fontaine et al. (2015) used chromosome-scale reference assemblies for six sibling species in that complex to resolve a contested species tree and to demonstrate pervasive autosomal introgression among species that nonetheless retain distinct vectorial capacity. The *An. gambiae* work and the present analysis differ in scale and in age: the *Anopheles* complex is older and its members more divergent (~5–15 % nucleotide divergence within the complex), whereas the *Cx. pipiens* complex is younger and more uniform (93–95 % ANI within ingroup; Section 3.5). The two complexes converge in showing that small-panel pangenomic analyses can expose introgression histories that single-locus or pairwise studies cannot. Broader context comes from Neafsey et al. (2015), who sequenced sixteen anopheline genomes spanning ~100 million years of divergence and reported elevated rates of gene gain and loss, gene shuffling and intron turnover in *Anopheles* relative to *Drosophila*. Our CAFE 5 results, which show 3,200–3,700 gene-family expansions per branch over a much shorter timescale (a few million years among the *Cx. pipiens* forms), suggest that elevated gene-family turnover continues into the recent evolutionary history of culicine mosquitoes and is not unique to *Anopheles*. Small-panel pangenome analyses using core/shell/cloud partitioning have a long history in bacterial and plant systems (Tettelin et al., 2005; Golicz et al., 2020; Bayer et al., 2020) but have not previously been applied to any mosquito species complex.

4.4 Transposable element landscape and genome-size conservation

The conservation of transposable element (TE) content across the four genomes (52.69–54.43 % interspersed repeats, Section 3.6) contrasts with the meaningful variation in predicted protein counts (23,089–24,531) and in pangenome compartment membership. TE content at this level is typical of culicine genomes more broadly (Arensburger et al., 2010), and its near-uniformity across the *Cx. pipiens* complex suggests that the most recent common ancestor of the complex already carried this repeat fraction and that no major TE expansions or purges have occurred during the divergences sampled here. The contrast between conserved TE proportion and variable protein content suggests that the gene-family dynamics quantified by CAFE 5 operate primarily within the gene complement rather than via large-scale repeat-driven genomic remodeling on the timescale of *Cx. pipiens* complex divergence.

4.5 Limitations and future directions

Three limitations bound the inferences that can be drawn from the present design. First, a single chromosome-scale assembly is currently available for each of the four ingroup forms, so within-form population-level variation is not captured; the cloud and shell compartments reported here represent between-form divergence rather than population polymorphism, and resequencing-based pangenome studies of any single form would likely expand the cloud compartment substantially. Second, the *Cx. tarsalis* outgroup, although now correctly identified and assembled, contributes only as an outgroup-only orthogroup donor in the pangenome partition; protein-mode BUSCO recovery is limited (40.3 %) because the annotation must be lifted from a divergent source, and *Cx. tarsalis* therefore does not enter the concatenation SCO phylogeny at the resolution required to root the ingroup at the species-tree level. Future inclusion of a native-annotated *Cx. tarsalis* assembly would tighten the species-tree rooting. Third, ANI and synteny were assessed across the four ingroup forms only in this analysis; including *Cx. tarsalis* on a common ANI scale is precluded by its divergence: sketch-based ANI (skani) returns no estimate for the *Cx. tarsalis*–ingroup pairs because identity falls below the method's reliable floor (Section 3.5), and the fragmented, contig-level *Cx. tarsalis* assembly also precludes the chromosome-scale synteny analysis applied to the ingroup. Quantifying outgroup divergence on a continuous scale would require an alignment-based identity estimate, which has no comparable lower bound; we leave this to future work.

The *Cx. pipiens* complex therefore joins the *An. gambiae* complex as one of the few medically important mosquito species complexes with a chromosome-scale comparative-genomic framework. Where the *An. gambiae* work revealed that vector-competence traits can be acquired through interspecific introgression and reshape transmission landscapes (Fontaine et al., 2015), the present analysis indicates that one widely recognized *Cx. pipiens* form (*Cx. p. pallens*) carries molecular signals consistent with a hybrid origin that earlier population-genetic markers had previously suggested but could not directly test at genome scale. Future work integrating population-level resequencing within each form, native *Cx. tarsalis* annotations, and explicit network-phylogenetic and *D*-statistic-based introgression tests (Hibbins and Hahn, 2022) will sharpen the inferences begun here.

5. Conclusions

We present the first pangenome analysis of the *Culex pipiens* species complex, integrating four chromosome-scale ingroup assemblies with a *Cx. tarsalis* outgroup under a single reproducible Snakemake workflow. OrthoFinder identified 16,568 orthogroups; restricted to the four ingroup forms, 92,821 genes (98.2 %) were partitioned into 11,284 core (71.1 %), 3,726 shell (23.5 %) and 871 cloud (5.5 %) orthogroups, with an additional 687 *Cx. tarsalis*-only orthogroups. Form-specific cloud orthogroups were significantly unevenly distributed across forms ($\chi^2 = 71.9$, $p = 1.7 \times 10^{-15}$), with *Cx. molestus* carrying the largest set. A concatenated 8,067-locus phylogeny resolved *Cx. pallens* and *Cx. quinquefasciatus* as sister taxa with full bootstrap support but only 58.9 % gene-tree and 45.1 % site concordance at the deep ingroup node. This discordance

pattern, combined with prior microsatellite and RADseq evidence (Fonseca et al., 2009; Aardema et al., 2020), is consistent with a hybrid origin of *Cx. pallens* from *Cx. pipiens* and *Cx. quinquefasciatus*. CAFE 5 detected 617 significantly evolving gene families with strong, lineage-wide expansion bias on every branch. Whole-genome ANI across the ingroup ranged 93.07–94.72 % (skani), with 1:1 chromosome-scale synteny and 258–294 candidate inversions $\geq$ 100 kb per pair. Transposable elements occupied a uniform fraction (52.69–54.43 %) of each genome. The resulting resource places the *Cx. pipiens* complex alongside the *An. gambiae* complex as one of the few medically important mosquito species complexes with a chromosome-scale comparative-genomic framework and provides a baseline against which future population-level resequencing of each form can be benchmarked.

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Author contributions

T.M.: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Project administration, Writing – original draft. E.M.: Writing – review & editing. T.Mi.: Writing – review & editing. K.K.: Writing – review & editing, Supervision. All authors read and approved the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All analyses are reproducible from the project repository at https://github.com/tylermaire/cx_pipiens_pangenome under a Conda-managed Snakemake workflow. Genome assemblies were retrieved from NCBI GenBank/RefSeq with the accessions listed in Table 1; the *Cx. tarsalis* assembly (CtarK1) was retrieved from the Open Science Framework at osf.io/mdwqx. All intermediate and final results files (orthogroup tables, alignments, phylogenetic outputs, CAFE outputs, ANI matrices and synteny coordinates) are bundled with the repository commit corresponding to this manuscript.

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Supplementary materials

Supplementary Table S1. BUSCO completeness (genome- and protein-mode) for all five assemblies. **Supplementary Table S2.** QUAST assembly statistics for all five assemblies. **Supplementary Table S3.** Pangenome partition summary (orthogroup and gene counts per

compartment). **Supplementary Table S4.** Per-form pangenome partition (orthogroup and gene counts per form per compartment). **Supplementary Table S5.** CAFE 5 per-lineage expansion/contraction counts with λ , α and significant family totals. **Supplementary Table S6.** Per-form copy numbers for detoxification, chemosensory and immunity gene categories. **Supplementary Table S7.** RepeatMasker repeat-class composition per genome. **Supplementary Table S8.** Distance to nearest transposable element per gene (median, quartiles, within-1 kb fraction), by pangenome compartment. **Supplementary Figure S1.** Per-pair gene-anchor synteny dotplots for the six ingroup genome comparisons, the per-pair basis for the composite Figure 6. **Supplementary Figure S2.** Overview of the end-to-end Snakemake analysis workflow, showing tools, key parameters and input/output files for each step (annotation transfer, orthology and pangenome partitioning, phylogenomics, gene-family evolution, ANI and synteny, repeat annotation, and figure generation).